25901831 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCAAT100 - LOGISTICS AND E-COMMERCE OPERATIONS II

(2025 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Kitting and assembly in warehouses mainly help in: [U] / [CO1]

   a). Reducing storage space

   b). Faster order fulfillment

   c). Increasing production cost

   d). Price control

   Answer b. Faster order fulfillment

2. Warehousing strategy mainly focuses on: [U] / [CO1]

   a). Advertising products

   b). Efficient storage and movement of goods

   c). Increasing product price

d). Market research

Answer b. Efficient storage and movement of goods

3. Warehousing mainly helps in: [U] / [CO1]

a). Production planning

b). Storage of goods until consumption

c). Advertising product

d). Price fixation

Answer b. Storage of goods until consumption

4. Mobile-based payments require: [U] / [CO4]

a). Smartphone or mobile device

b). Internet connectivity

c). Registered payment app or account

d). All of the above

Answer d. All of the above

5. One major benefit of TMS is: [U] / [CO2]

a). Faster delivery and cost reduction.

b). Increased paperwork.

c). Reduced shipment visibility.

d). Ignoring customer needs.

Answer a. Faster delivery and cost reduction.

6. Which tool is commonly used in IMS? [U] / [CO2]

a). Barcode scanning

b). Advertising posters

c). Paper invoice only

d). None of the above

Answer a. Barcode scanning

7. Which is not a function of WMS? [U] / [CO2]

a). Receiving goods

b). Storage management

c). Customer advertising

d). Order picking

Answer c. Customer advertising

8. E-cheques are processed via: [U] / [CO4]

a). Bank’s electronic system

b). Cash counter only

c). Manual ledger

d). ATM only

Answer a. Bank’s electronic system

9. Token-based systems help: [U] / [CO4]

a). Reduce fraud

b). Increase manual paperwork

c). Slow down payment processing

d). Replace cash with coins

Answer a. Reduce fraud

10. EDI primarily replaces: [U] / [CO5]

a). Manual, paper-based processes

b). Digital payments

c). Online advertising

d). E-banking

Answer a. Manual, paper-based processes

[1x10 = 10]

Part B

SHORT ESSAY

Answer any **3** questions

Each question carries **5** marks

11. What are the main functional key areas found in a warehouse? [U] / [CO1]

The key areas in a basic warehouse are as follows;

1. **Receiving Area**: This is the space for unloading, inspecting and sorting incoming goods.
2. **Storage Area**: The main area for holding inventory. The layout depends on the type of goods and storage methods and inventory turnover.
3. **Picking and Packing Area**: These are designated zones for retrieving items for orders and preparing them for shipment.
4. **Shipping Area**: This is the space for staging completed orders, loading onto outbound vehicles and documentation.
5. **Value-Added Services Area**: This is the area for kitting, labelling, assembly or other specialized tasks
6. **Office Space**: This is the space for administrative tasks and management. Ideally located to oversee operations.
7. **Support Areas**: This area includes break rooms, restrooms and equipment charging stations.

12. What is an Order Management System (OMS) and what are its main functions? [U] / [CO2]

An Order Management System (OMS) is a software solution that streamline and automates the entire life cycle of a customer order from the moment an order is placed until its final delivery and any post purchase activities like returns. It processes customer orders from placement to fulfilment.

Functions of OMS

1. **Order capture and entry**: It gathers orders from diverse sales channels, including e-commerce websites, market places (eg. Amazon, Flipkart in India), physical retail stores, call centers, mobile apps and B2B portals.
2. **Inventory Management and allocation**: Oms integrates with inventory management system and provides an accurate, unified view of stock level across all locations (warehouse, retail stores, in transit), preventing overselling or stock outs.
3. **Order Processing and Routing**: OMS manages the flow of orders through various stages like validation, payment processing, fraud checks, inventory allocation and fulfillment initiation. It facilitates

intelligent routing.

4. **Payment Processing**: OMS integrates payment gateways to authorize and capture payment securely. It facilitates the processing of refunds and handle chargeback disputes.

5. **Fulfilment and Shipping Co-ordination:** OMS transmits fulfilment requests to warehouse management system or 3 PL Partners. It helps to select the best carrier based on cost, delivery speed and service level.

6. **Return Management**: It manages the initiation of returns, issuing Return Merchandise Authorization (RMA). It tracks returned items inspects their condition, and initiate refund or exchange.

7. **Customer Service and Communication**: OMS allows customers to track their orders view order history and initiate returns independently. It sends automated updates to customers regarding order confirmation, shipment, delivery.

8. **Reporting and Analytics**: Oms tracks metrics such as order fulfilment rates, shipping costs, delivery times, returns rates and customer satisfaction. It provides data to improve demand forecasting and inventory planning.

13 Define LIS. Explain its components. [U] / [CO2]

Logistics Information System or LIS is a set of highly advanced digital programs running on integrated software infrastructure that helps business to plan and strategize to optimize their logistics operations and empowers them to effortlessly manage the operations such as procurement, storage, order picking, shipment tracking, transportation etc. it is designed to manage logistics operations efficiently. It enhances decision making, improves operational efficiency, supports supply chain management.

Components of LIS

1. **Data Management**: It is about capturing, storing, centralizing and accessing various types of transactional data related to inventory, orders shipments, warehouses activity, finance and suppliers.

2. **Order Management System (OMS):** An Order Management System (OMS) is a software solution that streamline and automates the entire life cycle of a customer order from the moment an order is placed until its final delivery and any post purchase activities like returns. It processes customer orders from placement to fulfilment.

3. **Transportation Management System (TMS):** This is a software application that help to plan, execute and optimize the movement of goods. It facilitates route planning, carrier selection, freight auditing.

4. **Inventory Management System (IMS):** This is a software tools that tracks inventory levels, orders, and deliveries. IMS helps to maintains optimal stock levels and reduces excess inventory.

5. **Warehouse Management System (WMS):** This is a software that manages warehouses operations including inventory control, order fulfilment and shipping. It optimizes space utilization and enhances picking and packing process.

6. **Data Analytics and Reporting Tools:** These are the tools that provide insights into logistics performance through data analysis. It supports informed decision making.

14. What are the major risks in E- Payment? [An] / [CO4]

The various risks in e-payments are listed below:

1. **Risks from Fraud or mistake:** There is a chance for risk of frauds in e-payments. This may arise due to inadequate security mechanisms and controls.
2. **Privacy Risks:** Data protection risk. The personal as well as payment information given for making electronic payments may be leaked or misused if the data protection mechanisms are weak.
3. **Security Risks**: Data Reliability risk. This is related to the authentication of parties involved. Security risk arises on account of unauthorized access to a critical information stores like customer data bases, credit card details etc.
4. **Credit Risks**: Credit risk is a major concern in net settlement systems. This happens because a party's failure to settle its payment obligation, due to insufficiency of funds.
5. **Operational Risks:** Operational risk can be defined as the risk of loss resulting from inadequate or failed internal processes, people, and systems or from external events

15. Define ERP and describe its components in E-commerce.

ERP software digitally records every business transaction a company makes from the issuance of a purchase order to the consumption of inventory. Theses are multi module application that includes financial management, business intelligence, supply chain management, HR Management, customer service, production planning etc. [U] / [CO5]

ERP Components

1. **Data Warehouse**: A database that collects business information from many sources in the enterprise, covering all aspects of the company's processes, products etc.

2. **Legacy Systems**: Legacy systems are the old or existing computer systems and software that a company was using before implementing an ERP system. In order for an ERP software deployment to be useful it needs to be integrated with the legacy systems the organizations uses.

3. **On-Line Analytical Processing (OLAP):** This component of ERP helps the users to analyze scenarios and take inferences and decisions using up to date data with the help of business, econometric and mathematical models.

4. **Bolt -on Application:** A bolt-on application is an additional software that is attached to an existing ERP system to add extra features or functions. This includes the specification, rules and standard operating procedures pertaining to the industry (Manufacturing, Services and Retailing etc.) in which the business operates.

5. **Operational Databases**: This includes databases of different customers, production process, vendors/suppliers, inventory etc. This database will provide necessary information to the various business process like procurement, manufacturing, inventory management etc.

[5x3 = 15]

Part C

ESSAY

Answer any **1** questions

Each question carries **10** marks

16. Explain Online Debit and Credit Card based payment systems?

Debit card is a plastic card that provides the card holder electronic access to his or her bank accounts through ATM. Debit cards are Pay Now cards, the payments made using debit cards are immediately debited from the card holder designated bank account. VISA, MasterCard, Maestro and RuPay are the major debit card brands. [U] / [CO4]

Credit card is defined as a plastic card authorizing a purchase on credit. This card enables the card holder to purchase goods, travel or dine without making immediate payments. The credit card issuing bank makes payment to the merchants (from the credit granted to the cardholder) concern and sends statement to the card holder periodically.

Debit Card	Credit Card
It is a pay now product	It is pay later product
Payment is immediately debited in card holders account	The card holder can avail credit for 30 to 45 days
No interest charges on the use of debit card	Interest and other charges have to be paid on credit card
Probability of risk of fraud is lower	Probability of risk of fraud is higher
No risk of overspending as the customer can spend what he has	The holder tends to overspending because he is using the credit provided by credit card issuer
It can be used at ATM terminal to withdraw cash	It is commonly used to make payments at POS/online. it can be used in ATM with restricting (like withdrawal amount, high charges)
A bank account (Savings /Current account) and minimum balance should be maintained	Minimum balance in account is not compulsory in case of credit cards

17. Define ERP systems in Logistics. Analyze the benefits of ERP. [An] / [CO2]

ERP in logistics is a software system that helps a company manage and track the movement of goods from purchase to delivery in an organized and efficient way.

An Enterprise Resource Planning (ERP) system is a comprehensive software suite that integrates and manages an

organization’s core business processes across various departments into a single, unified system. It provides a centralized database and unified view of operations, improving efficiency and enabling data-driven decision-making. ERP systems also facilitate communication with external stakeholders such as suppliers and customers.

Examples of ERP Solutions

SAP, Oracle ERP Cloud, Microsoft Dynamics 365, Tally ERP, Zoho ERP

1. Single Source of Truth (Centralized Database)

In logistics, many departments like procurement, warehousing**,** transportation, and sales use data.

- ERP keeps all data in one central system
- Everyone sees the same updated information
- If stock is updated in the warehouse, the sales and transport teams can immediately see it.
  This avoids confusion like sending goods that are already out of stock.

2. End-to-End Visibility

ERP gives a complete view of the entire logistics process from start to finish.

- From ordering raw materials → storing → shipping → delivery
- Managers can track goods at every stage

We can know:

- Where the goods are (warehouse / in transit)
- Delivery status
- Delays or bottlenecks
- Helps in better decision-making and faster problem-solving.

3. Process Automation

ERP automates repetitive logistics tasks.

- Automatic order processing
- Inventory updates without manual entry
- Auto-generation of invoices and shipping documents

When a customer places an order:

- ERP automatically checks stock
- Creates delivery order

- Updates inventory
- Reduces human errors and saves time.

4. Improved Collaboration

ERP connects all departments in logistics.

- Procurement, warehouse, finance, and transport teams work together smoothly
- Information flows easily between departments

If delivery is delayed:

- Warehouse, transport, and customer service teams are all informed instantly
- Results in better coordination and customer satisfaction.

[10x1 = 10]